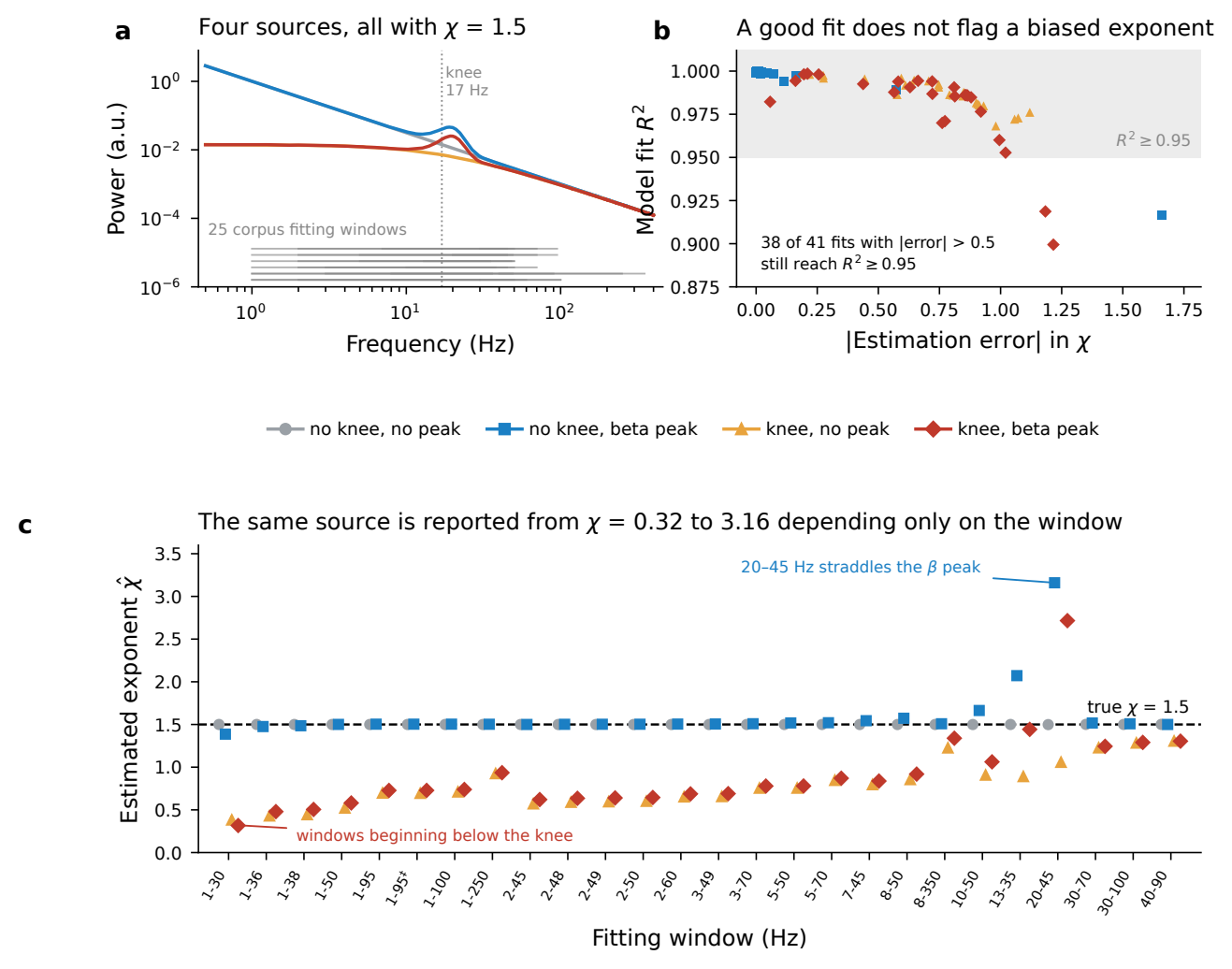
a
Four sources, all with $\chi$ = 1.5
knee
17 Hz
25 corpus fitting windows
Power (a.u.)
Frequency (Hz)
b
A good fit does not flag a biased exponent
$R^2 \geq 0.95$
38 of 41 fits with |error| > 0.5
still reach $R^2 \geq 0.95$
Model fit $R^2$
|Estimation error| in $\chi$
no knee, no peak
no knee, beta peak
knee, no peak
knee, beta peak
c
The same source is reported from $\chi$ = 0.32 to 3.16 depending only on the window
20–45 Hz straddles the $\beta$ peak
true $\chi$ = 1.5
windows beginning below the knee
Estimated exponent $\hat{\chi}$
Fitting window (Hz)
1-30
1-36
1-38
1-50
1-95
1-95*
1-100
1-250
2-45
2-48
2-49
2-50
2-60
3-49
3-70
5-50
5-70
7-45
8-50
8-350
10-50
13-35
20-45
30-70
30-100
40-90